

Chapter 2: Consumer Theory

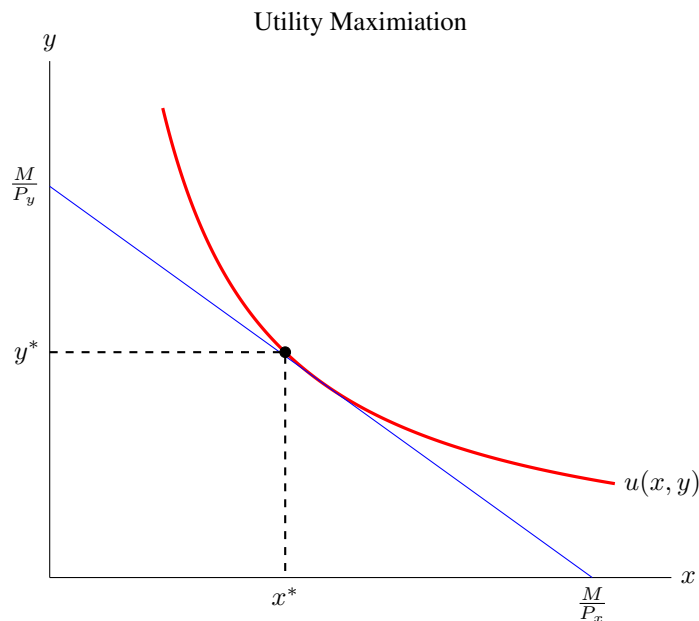
1 Indifference Curves and the Budget Constraint

This chapter goes over the concepts discussed in Block 4 and 9 of EC1002. The main theme is that we develop a theory of how individuals behave in a hypothetical economy.

The principle objective function that an individual solves is to maximise their satisfaction (utility function) subject to their budget constraints:

$$\max u(x, y) \quad \text{s.t.} \quad P_x x + P_y y = M$$

We learnt that this leads to the graphical solution:



That is, the optimal point that maximises utility is when the slope of the indifference curve (also called the marginal rate of substitution) is equal to the slope of the budget constraint:

$$MRS = \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

In this module, we need to derive this result mathematically. The starting point is the objective function and its constraint:

$$\max U(x, y) \quad \text{s.t.} \quad p_x x + p_y y = M$$

To solve this, we need to use the Lagrangian:

$$\mathcal{L} = u(x, y) + \lambda(M - P_x x - P_y y)$$

Taking the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial u}{\partial x} - \lambda P_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{\partial u}{\partial y} - \lambda P_y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - P_x x - P_y y = 0.$$

From the first two conditions:

$$\frac{\partial u}{\partial x} \frac{1}{P_x} = \frac{\partial u}{\partial y} \frac{1}{P_y},$$

$$\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = \frac{P_x}{P_y}$$

$$\frac{MU_x}{MU_y} = MRS = \frac{P_x}{P_y},$$

which is the tangency condition that leads to the graphical solution.

For an explicit utility function, its possible to derive the optimal values of x and y when subject to the budget constraint.

Example 2.1: A consumer has the following Cobb–Douglas utility function:

$$u(x, y) = x^{1/2}y^{1/2}$$

where $\alpha, \beta > 0$. The price of x is P_x and the price of y is P_y . The consumer's income is M . Derive the demand functions for x and y .

Solution: The consumer's problem is:

$$\max_{x, y} x^{1/2}y^{1/2} \quad \text{s.t.} \quad P_x x + P_y y = M$$

Using the Lagrange multipliers method:

$$\mathcal{L} = x^{1/2}y^{1/2} + \lambda(M - P_x x - P_y y)$$

Deriving the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{x^{-1/2}y^{1/2}}{2} - \lambda P_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{x^{1/2}y^{-1/2}}{2} - \lambda P_y = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - P_x x - P_y y = 0.$$

From the first two equations, we start by making y the subject:

$$\frac{x^{-1/2}y^{1/2}}{2P_x} = \frac{x^{1/2}y^{-1/2}}{2P_y}$$

$$\frac{y^{1/2}}{y^{-1/2}} = \frac{2P_x}{2P_y} \frac{x^{1/2}}{x^{-1/2}}$$

$$y = \frac{P_x}{P_y} x$$

To find the optimised value of x , we substitute y into the budget constraint and make x the subject:

$$M = P_x x + P_y y$$

$$M = P_x x + P_y \frac{P_x}{P_y} x$$

$$M = P_x x + P_x x$$

$$x^* = \frac{M}{2P_x}.$$

The optimised value of y is therefore:

$$y = \frac{P_x}{P_y} x$$

$$y = \frac{P_x}{P_y} \frac{M}{2P_x}$$

$$y^* = \frac{M}{2P_y}$$

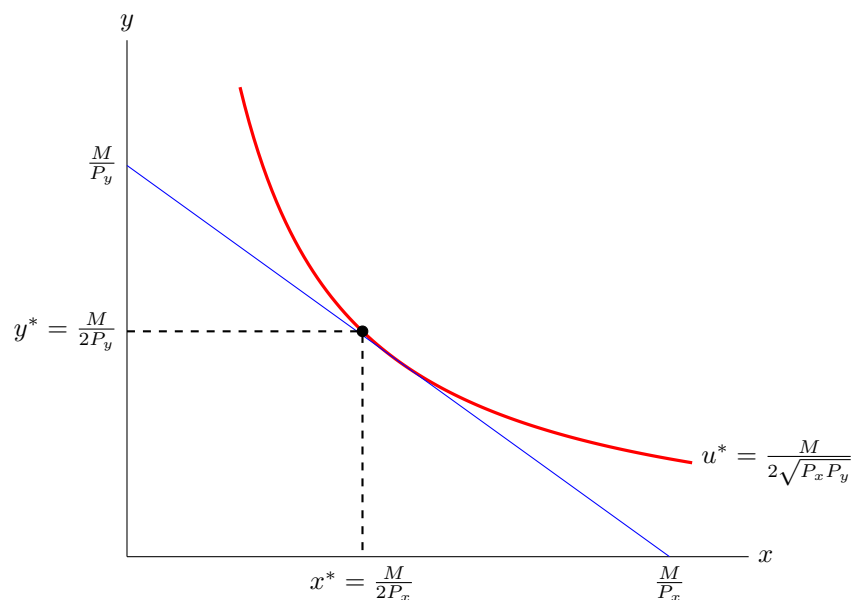
Which means that the value of utility at the optimum is:

$$u^* = x^{1/2} y^{1/2}$$

$$= \left(\frac{M}{2P_x} \right)^{1/2} \left(\frac{M}{2P_y} \right)^{1/2}$$

$$= \frac{M}{2\sqrt{P_x P_y}}$$

Graphically, this can be shown as:



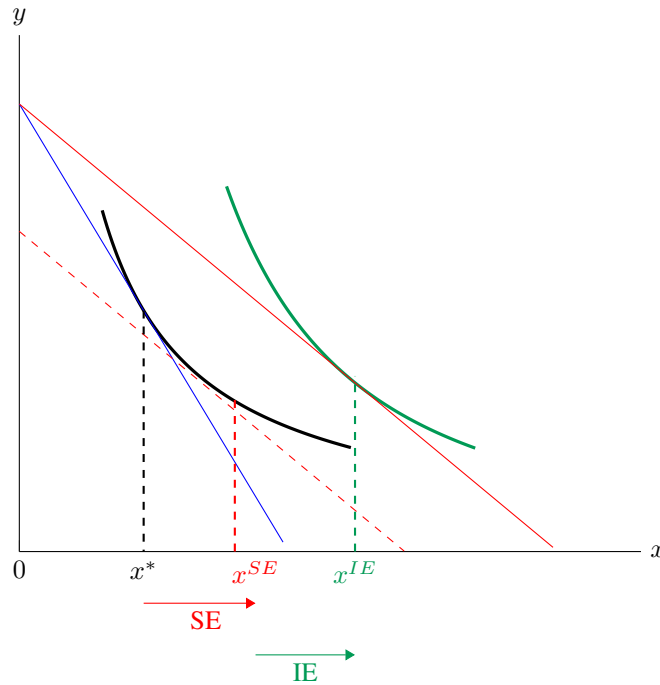
2 Adjustment to Prices

We discussed that when a price of a good changes, then a consumer theoretically chooses a new consumption bundle that incorporates the substitution and income effect. In EC1002, this was calculated using the Hicksian decomposition. In this module, we demonstrate this method mathematically.

2.1 The Hicksian Decomposition

Following a change in price, the Hicksian decomposition asks "how many goods would the consumer purchase at the new price but at their old utility curve".

For a normal good, we can show this graphically as:



Let us now derive this result mathematically:

Example 2.2: Suppose $u(x, y) = x^{1/2}y^{1/2}$. Income is at $M = 72$, the price of y is set at $P_y = 1$ and the price of x is initially set at $P_x = 9$ but reduces to $P'_x = 4$. Calculate the total price effect of x using the Hicksian decomposition.

Solution:

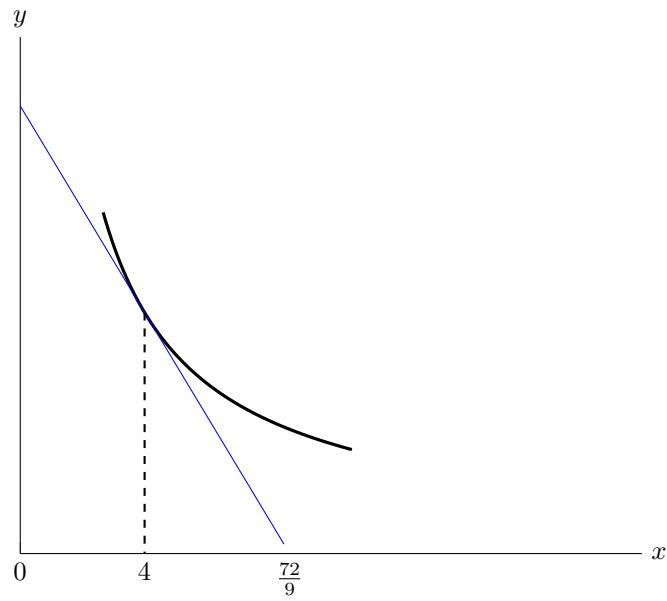
For the utility function $u(x, y) = x^{1/2}y^{1/2}$ and budget constraint $P_x x + P_y y = M$, we have derived the solution in Example 2.1. To keep things simple, we only focus on x :

$$x^* = \frac{M}{2P_x} = \frac{72}{2 \cdot 9} = 4$$

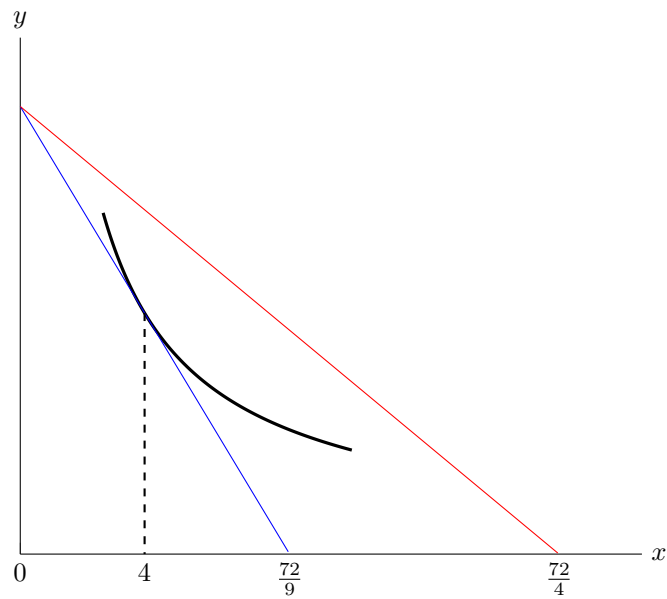
where the optimal utility is a constant given as:

$$u^* = \frac{M}{2\sqrt{P_x P_y}} = \frac{72}{2\sqrt{9 \cdot 1}} = 12$$

Graphically, this would be shown as:

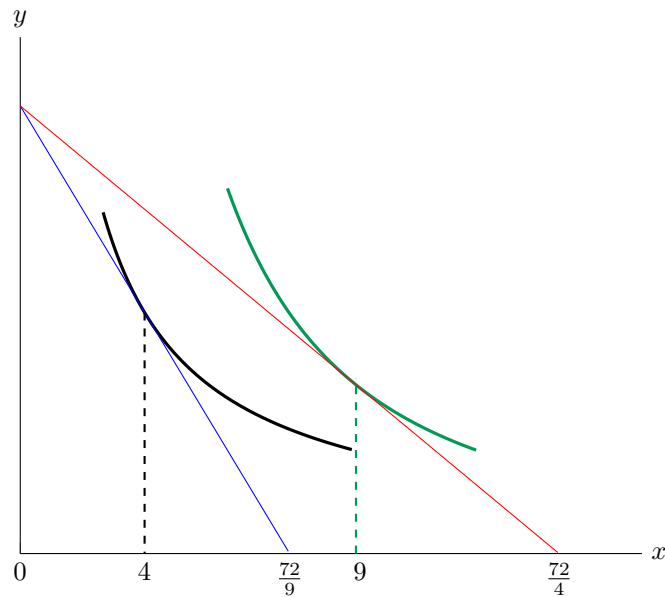


Following the price decrease to $P'_x = 4$, we can calculate the new budget constraint as $\frac{M}{P'_x} = \frac{72}{4} = 18$:



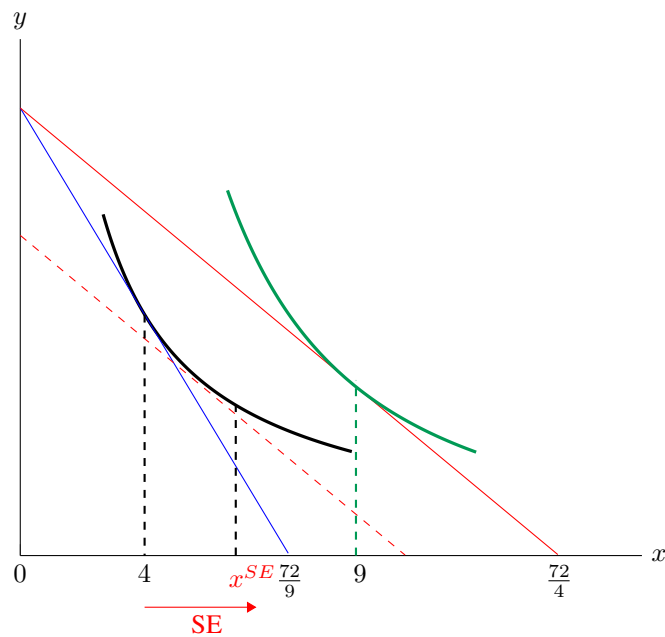
At the new price level, the consumer will optimally select:

$$x'^* = \frac{M}{2P'_x} = \frac{72}{2 \cdot 4} = 9$$



This means that the total price effect (PE) is $9 - 4 = 5$. However, the 9 includes both income and substitution effects. We need a way to decompose it from this combined amount.

We calculate the substitution effect using the Hicksian method. Graphically, we draw a budget line that has the same slope as the new budget line, but make it tangent to the old utility curve:



Mathematically, we know that the optimal point is when $x^* = \frac{M}{2P_x}$. This means that x^{SE} can be described as:

$$x^{SE} = \frac{M - \Delta M}{2P'_x},$$

where the ΔM component is called the **compensating variation**. It is defined as the amount of money that we need to take away^a in order to bring the consumer back to the original utility curve. In order to derive the explicit value of ΔM , you need to note that the optimal utility at the substitution effect is:

$$u^{SE*} = \frac{M - \Delta M}{2\sqrt{P_x P_y}}$$

However, according to the Hicksian decomposition, we know that $u^{SE*} = u^*$. That is, they are both on the same utility curve (you can see this in the graph). Therefore, we can use this result to derive the compensating variation ΔM :

$$\frac{M - \Delta M}{2\sqrt{P'_x P_y}} = u^*$$

$$\frac{M - \Delta M}{2\sqrt{P'_x P_y}} = \frac{M}{2\sqrt{P_x P_y}}$$

$$\frac{72 - \Delta M}{2\sqrt{4 \cdot 1}} = \frac{72}{2\sqrt{9 \cdot 1}}$$

$$\Delta M = 24$$

Substituting this into x^{SE} :

$$\begin{aligned} x^{SE} &= \frac{M - \Delta M}{2P'_x} \\ &= \frac{72 - 24}{2 \cdot 4} = 6 \end{aligned}$$

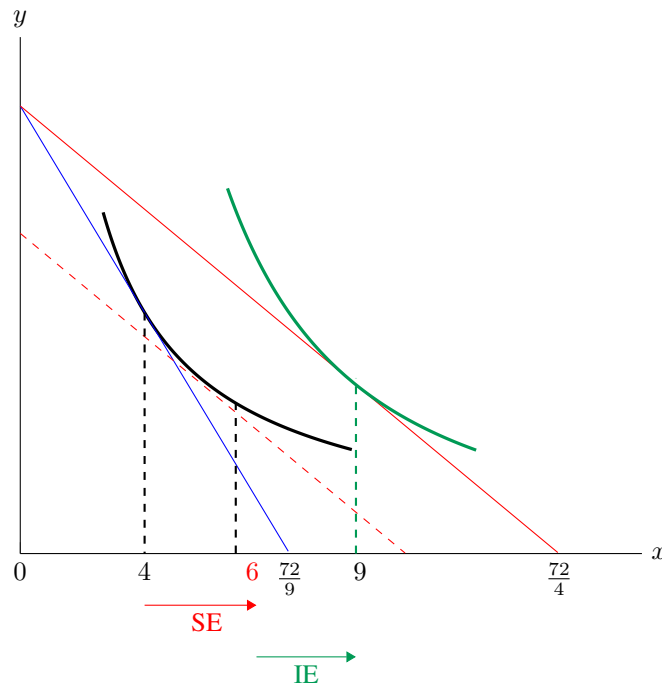
We can therefore decompose the price effect from the income and substitution effect:

$$PE = (SE + IE) - 4 = 5$$

$$= (6 + IE) - 4 = 5$$

$$IE = 3$$

Graphically:



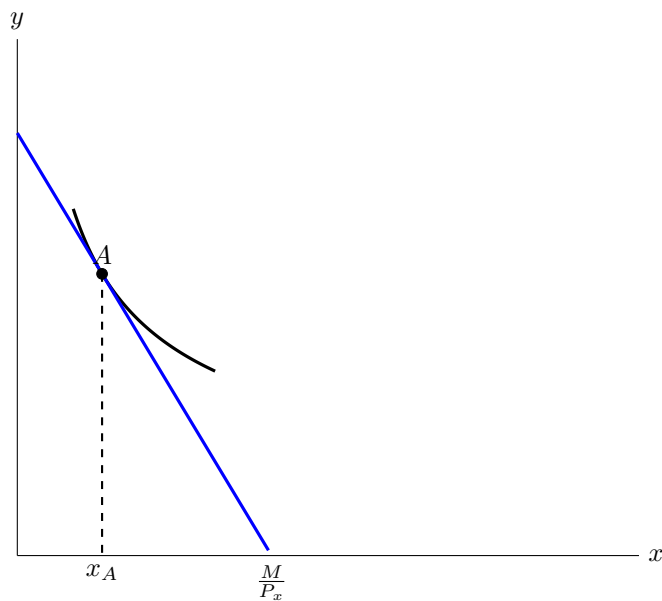
^aIt is also the amount of money that we *give* in the instance of a price increase.

2.2 The Equivalent Variation

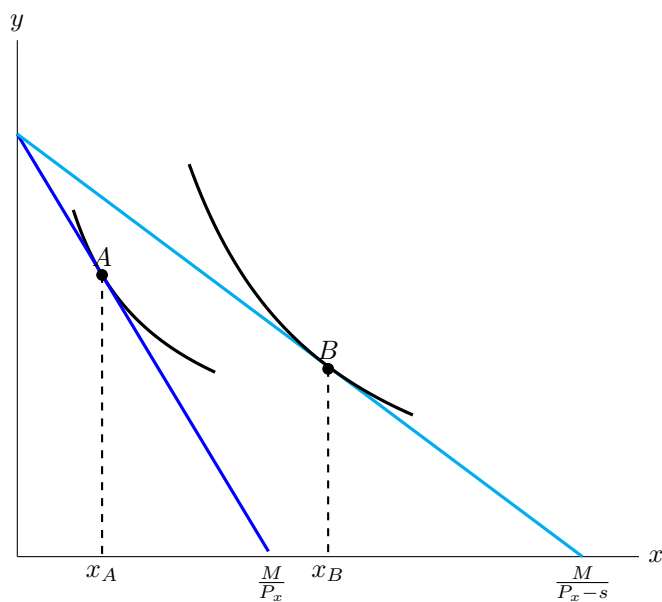
In the previous section, we introduced the concept of compensating variation. Following a price adjustment, this is the **amount of income that we would change** from the consumer in order to restore them back to the **original utility function**, but at the **new price ratio**.

A related concept in microeconomic theory is called the equivalent variation, which is essentially the reverse. Following a price adjustment, this is the **amount of money that we would change** from the consumer in order to keep them at the **new utility curve**, but at the **old price ratio**.

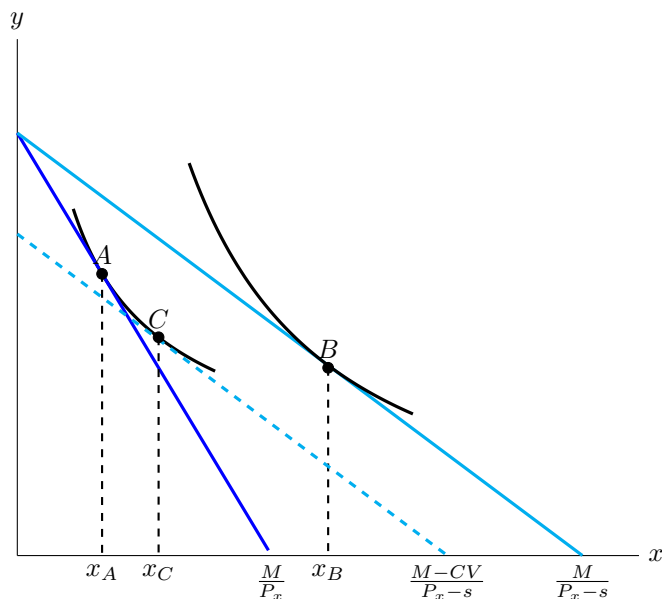
There are important policy considerations based on how we model behaviour in an economy. Consider an individual who has a budget M set for spending between food and fuel. For a given utility function, we can illustrate their optimal choice as:



Now suppose that the government wants to intervene and offer a subsidy s for fuel to consumers. This policy would have the same effect of a price decrease of fuel $P_x - s$:



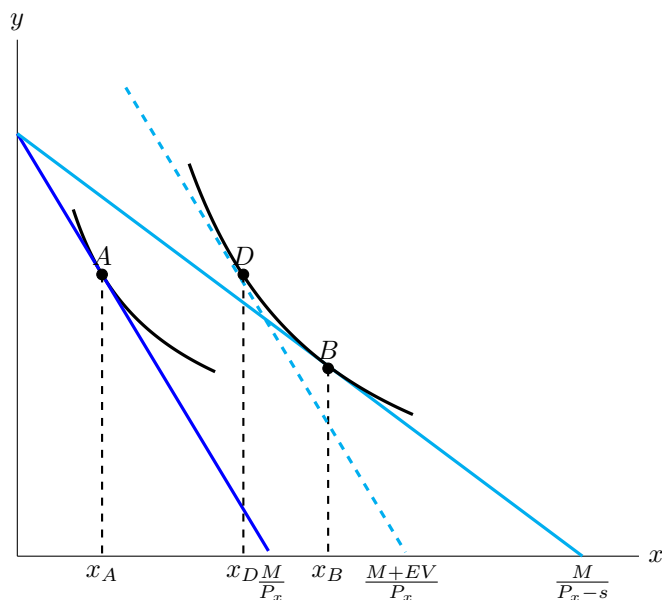
The total price effect would be the movement from A to B . If we were to illustrate the substitution effect using the Hicksian decomposition, we would calculate the compensating variation to restore the consumer to the original utility curve but at the new price ratio:



Overall, we can clearly see that the subsidy policy is good for the consumer. This is because they will choose the point B which is on a higher utility curve than on A .

It is interesting to note that the government is currently spending $s \cdot x_B$ amount of money in order to implement the policy. This raises an important question as to whether we can achieve the *same outcome* of consumer satisfaction but at a lower cost to the government.

Consider that instead of giving consumers a subsidy s (so the price of heating is back to P_x), the government transfers a lump-sum amount of money that keeps the consumer on *the new utility curve*:



Notice that even though at D the individual is consuming less x compared to point B they are still on the same level of utility as the subsidy policy. The lump-sum amount of money that achieves this specific outcome is known as the equivalent variation.

The policy of transferring an equivalent variation is cheaper on the government compared to the subsidy option. One

way to see this is that if the second option was implemented, then point B is no longer feasible as it geometrically on a higher budget constraint.

The intuitive difference between the two policies is that the subsidy creates a substitution effect on the consumer's decision making. The fact that only fuel is cheaper would mean the consumer ends up purchasing more of it^a. On the other hand, an equivalent variation in income would lead the consumer to more equally distribute their spending behaviour between both x and y .

An important point to keep in mind is that when the price of a good **decreases**, it means that the compensating variation **deducts** from the income, whereas the equivalent variation **adds** to the income. You must keep in mind that if the price of a good **increases**, then the signs will reverse. The below example highlights this.

Example 2.6: Suppose a consumer has the utility function $u(x, y) = x^{1/2}y^{1/2}$. For a budget constraint $M = P_x x + P_y y$, we know that this leads to the consumption points:

$$x^* = \frac{M}{2P_x} \quad y^* = \frac{M}{2P_y}$$

Which leads to the optimal utility level:

$$u^* = \frac{M}{2\sqrt{P_x P_y}}$$

Let us say that $M = 100$, with the price of y $P_y = 1$ and the price of x increases from $P_x = 1$ to $P'_x = 2$.

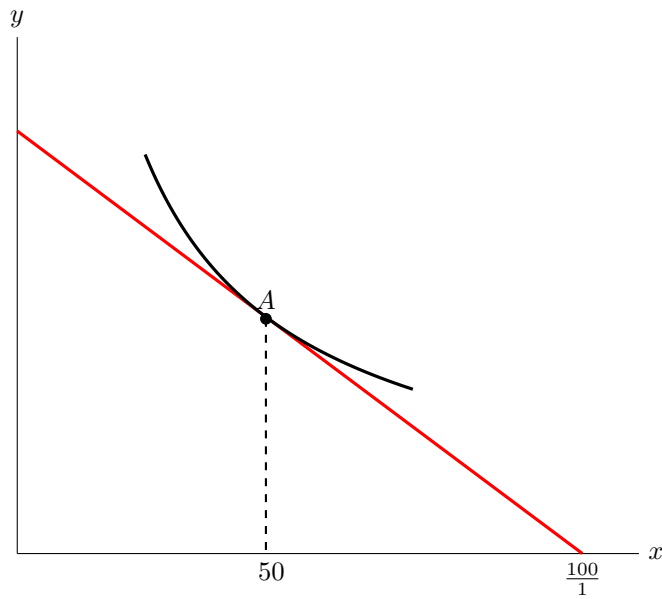
Focusing just on good x , calculate and show graphically the Hicksian decomposition (which clearly shows the compensating variation amount). After that, calculate and illustrate a policy of equivalent variation following the price increase.

Solution:

Let us calculate the initial point of x^A . This is simply:

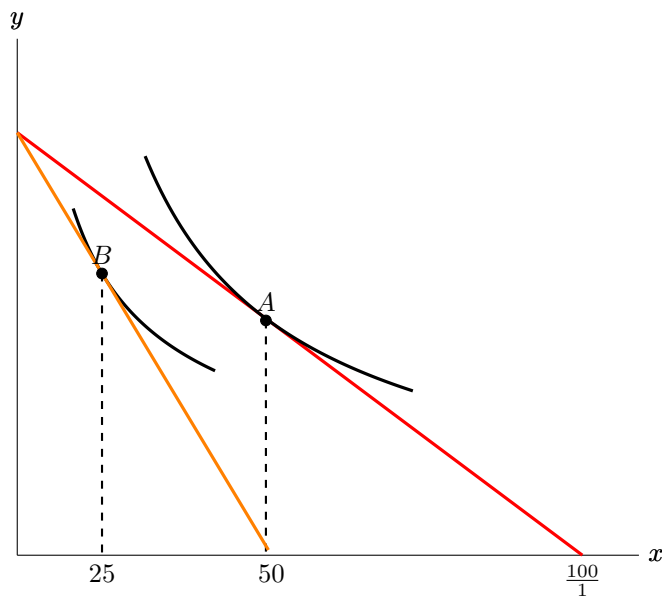
$$x^A = \frac{M}{2P_x} = \frac{100}{2} = 50$$

^aConsider a subsidy on electricity. This usually leads to individuals becoming less careful about the amount of electricity they use in their homes. It ultimately leads to a greater increase in usage and increases the costs on the government.



Following the price increase, the budget line rotates inwards. The new optimal point is:

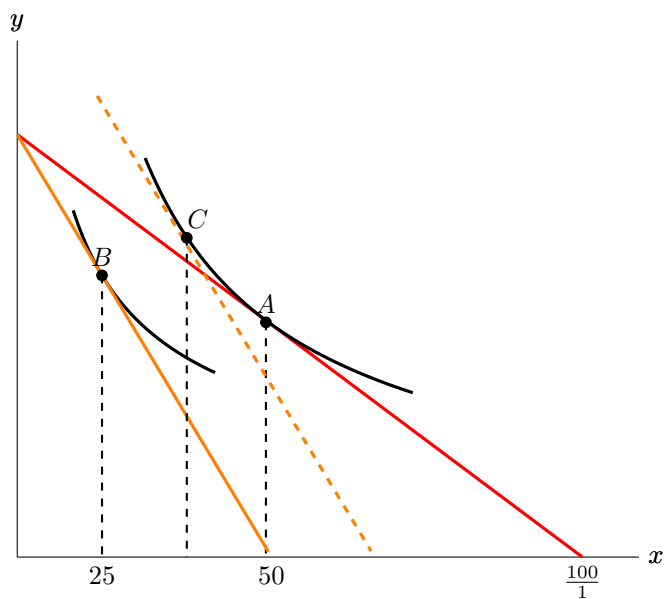
$$x^B = \frac{M}{2P'_x} = \frac{100}{2 \cdot 2} = 25$$



Therefore, the movement from point A to B represents the total price effect

We now need to calculate the substitution effect using the Hicksian decomposition. That is, we want to calculate the level of consumption at the original utility curve but with the new price ratio:

$$x^C = \frac{M + CV}{2P'_x}$$



We do not have the compensating variation value $CV = \Delta M$ just yet (notice that this is an addition given that we are illustrating a price *increase*). Since the original bundle A and substitution amount C are in the same utility curve, then we can say that:

$$u^C = u^A$$

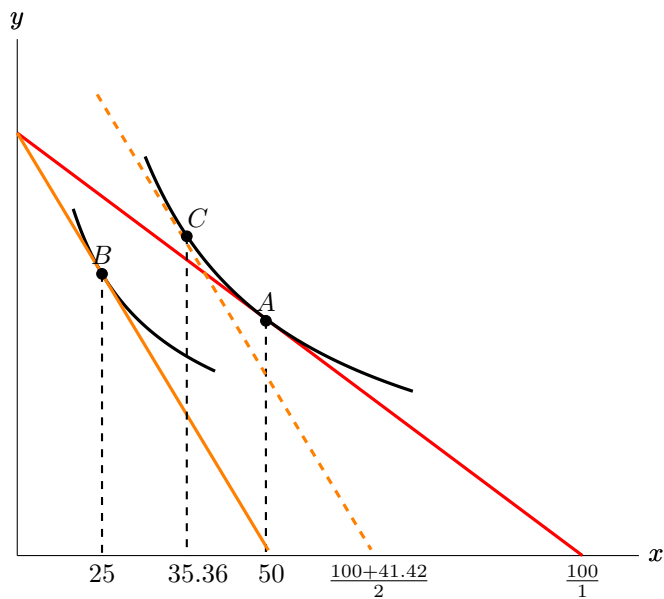
$$\frac{M + CV}{2\sqrt{P'_x P_y}} = \frac{M}{2\sqrt{P_x P_y}}$$

$$\frac{100 + CV}{2 \cdot \sqrt{2 \cdot 1}} = \frac{100}{2 \cdot \sqrt{1 \cdot 1}}$$

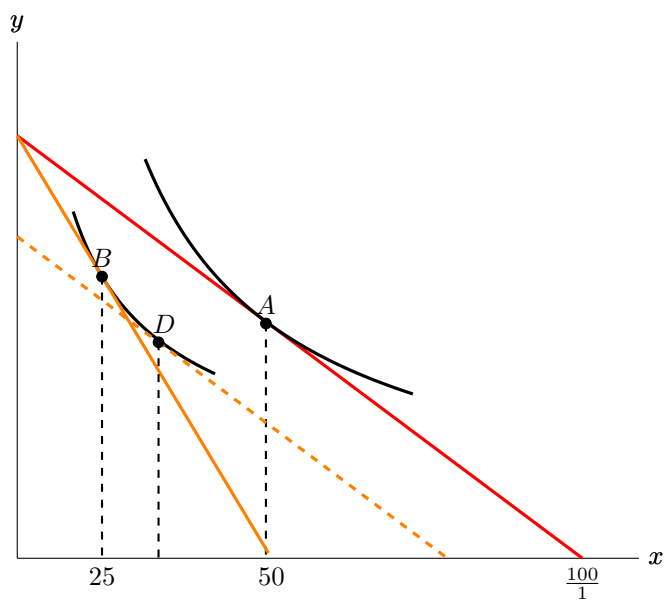
$$CV = 41.42$$

Where $CV = 41.42$ is the compensating variation. This means that x^C can be calculated as

$$\begin{aligned} x^C &= \frac{M + CV}{2P'_x} \\ &= \frac{100 + 41.42}{2 \cdot 2} = 35.36 \end{aligned}$$



Now let us consider a policy of equivalent variation. That is, we want to keep the consumer at the new utility curve but using the old price ratio:



We can describe point the consumption of point D as:

$$x^D = \frac{M - EV}{2P_x}$$

Just as before, we need to derive the equivalent variation value. To do this, notice that B and D are in the same utility curve. This means that:

$$u^B = u^D$$

$$\frac{M}{2\sqrt{P_x^I P_y}} = \frac{M - EV}{2\sqrt{P_x P_y}}$$

$$\frac{100}{2\sqrt{2 \cdot 1}} = \frac{100 - EV}{2\sqrt{1 \cdot 1}}$$

$$EV = 29.29$$

3 The Intertemporal Consumption Choice

Most of the economic analysis conducted at the introductory level is static. This means that we only observe equilibrium changes in a single time period. In order to examine higher level models, we can extend our analysis dynamically across time.

3.1 Model Setup

Consider an economy which exists only for two time periods $t = 1, 2$. In the first period, an individual earns Y_1 but also spends C_1 . We can define the difference as:

$$Y_1 - C_1 = S$$

We can see that if $Y_1 > C_1$, then the individual will have money saved that they can use in the next period. Alternatively, if $Y_1 < C_1$, then they will need to borrow money from the bank.

If the individual has a positive S , then the bank will pay them $(1 + r)$ on the savings in the next period. If they have a negative S , then the individual needs to make sure that it is paid back with interest $(1 + r)$ the next period.

Now in the second period, the consumer also earns Y_2 and must also decide on how much to spend C_2 . The intertemporal budget constraint can therefore be defined as:

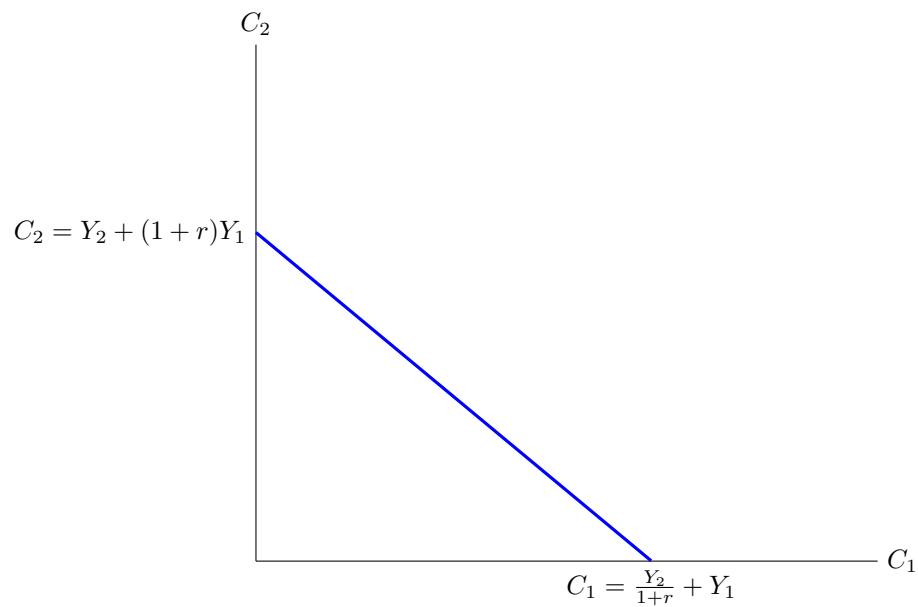
$$C_2 = Y_2 + (1 + r)S$$

$$C_2 = Y_2 + (1 + r)(Y_1 - C_1)$$

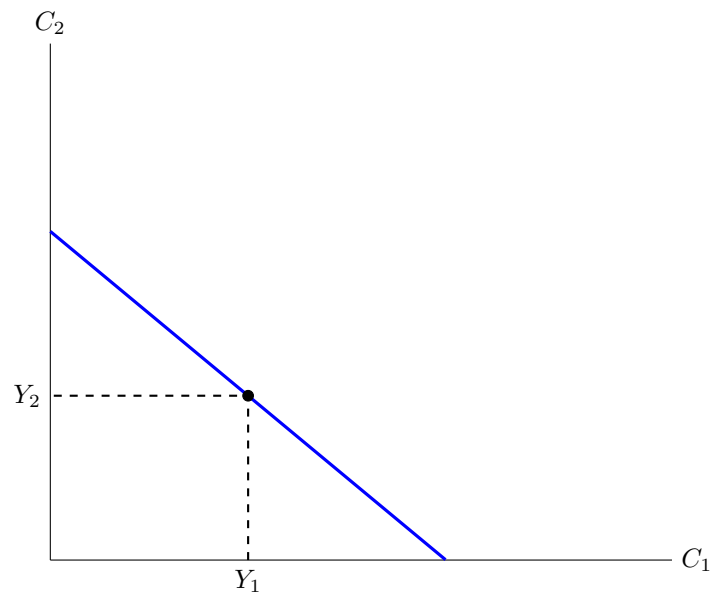
$$C_2 = Y_2 + (1 + r)Y_1 - (1 + r)C_1$$

Let us plot C_2 on the vertical axis and C_1 on the horizontal axis. This is a downward sloping linear equation:

$$\frac{\partial C_2}{\partial C_1} = -(1 + r)$$



We can illustrate the income endowments Y_1 and Y_2 at:



Now we need to translate how a consumer decides to become a saver or a borrower. Essentially, this is a decision to depends on each individual. We can think of their choice of C_1 and C_2 in the same way as they decide between two goods x and y .

Formally, the consumer maximises their utility function of their choice (C_1, C_2) subject to their intertemporal budget constraint:

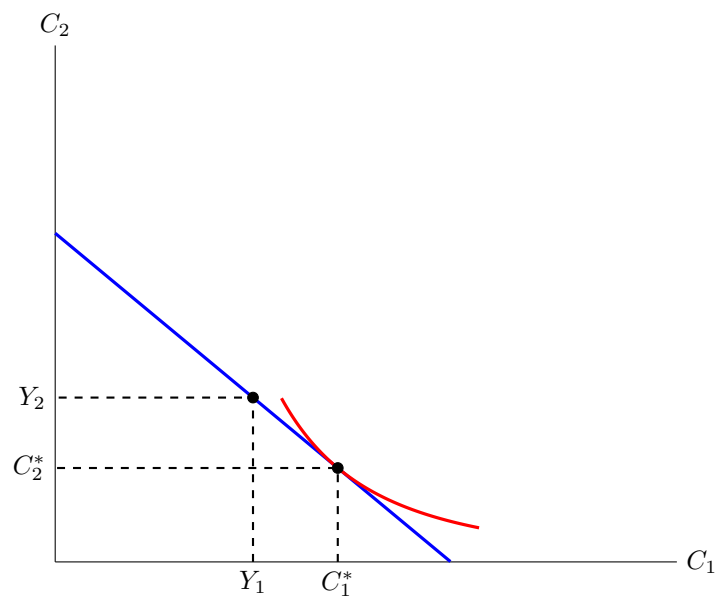
$$\max u(C_1, C_2) \quad \text{s.t.} \quad C_2 = Y_2 + (1+r)Y_1 - (1+r)C_1$$

Using the Lagrangian, we arrive at an identical solution:

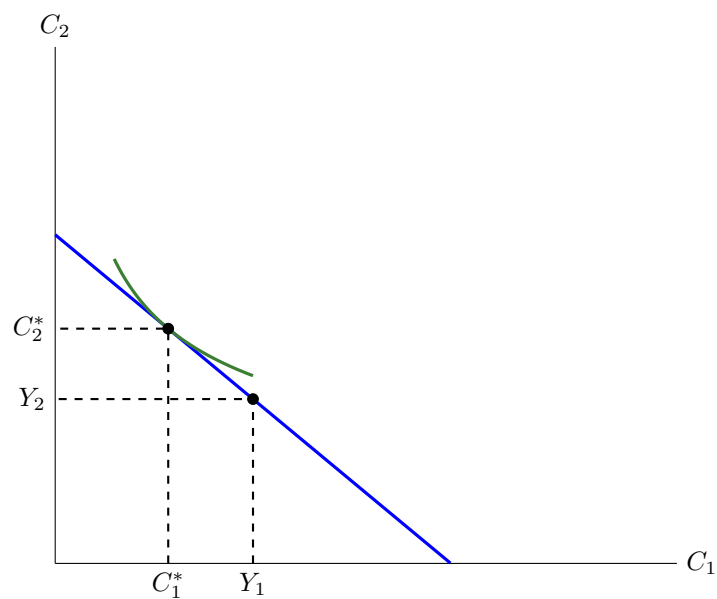
$$\frac{MU_{C_1}}{MU_{C_2}} = 1 + r,$$

which means that the consumer selects a bundle when their utility curve is tangent to their budget constraint.

If the individual was a borrower, then they will choose a consumption combination where C_1 exceeds their first period income Y_1 :



Similarly, if the consumer was a saver, then they select a consumption combination where C_1 is less than their first period income Y_1 :

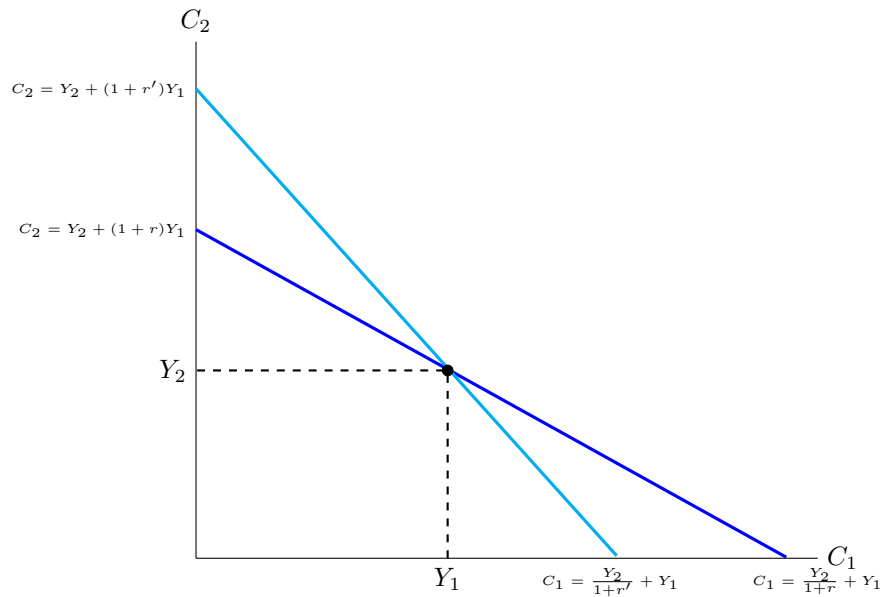


3.2 Adjustment to Interest Rate

Our analysis of the intertemporal consumption choice is identical to that of a consumer choosing between goods x and y . We saw that when the price of good x changes, the budget line rotates due to the horizontal intercept shifting.

In intertemporal case, the price of consumption is the interest rate r and *both* intercept points share this variable. Therefore, a change in interest rate *pivots* in both directions.

For example, suppose the interest rate increases $r' > r$. This can be shown as:

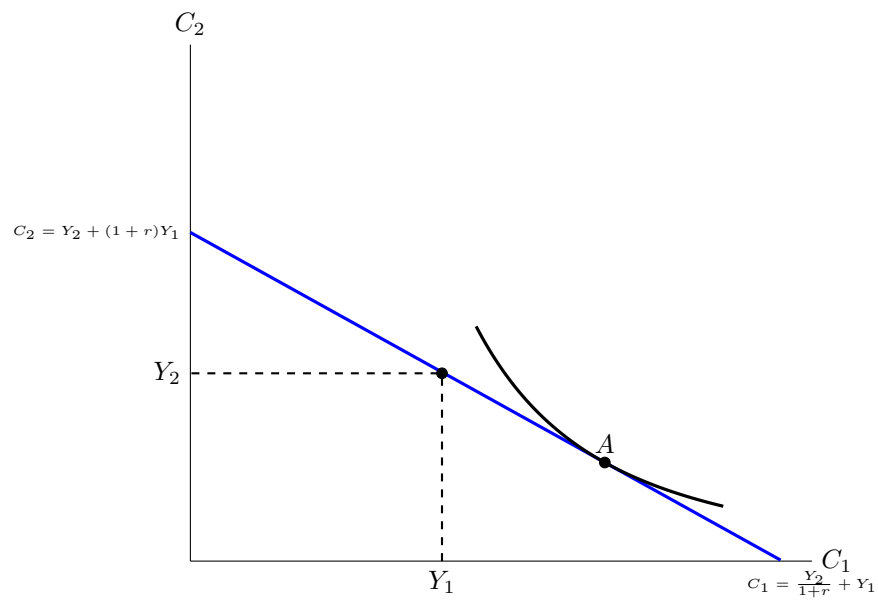


Notice how the new budget line passes through the initial income position (Y_1, Y_2) .

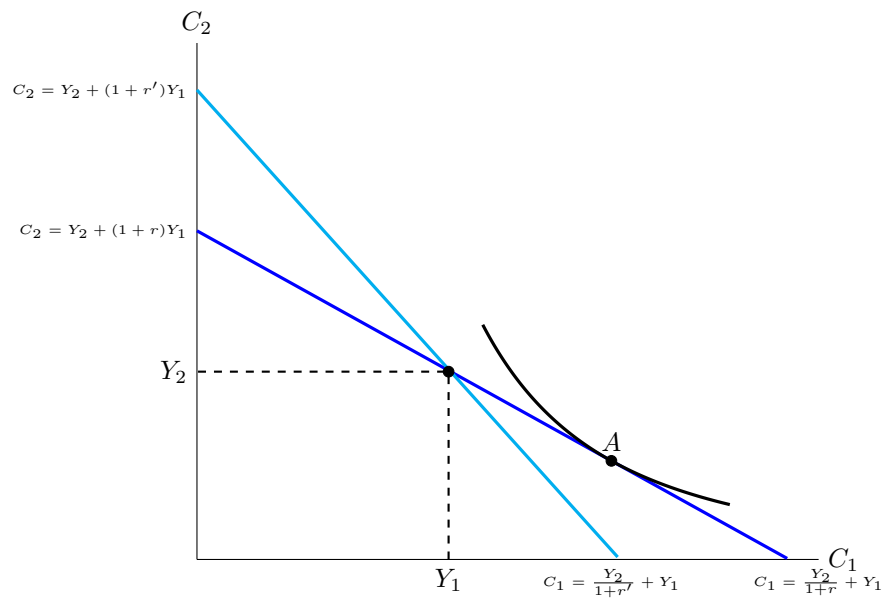
Just like in the case of goods x and y , a change in the interest rate leads to a substitution and income effect on the new consumption choice. However, we will see that the outcome differs based on whether the individual is a borrower or a saver.

Example 2.A: Consider an individual making an intertemporal consumption decision. Illustrate their initial choice given that they are a borrower. Demonstrate graphically the substitution and income effect that follows from an increase in interest rate.

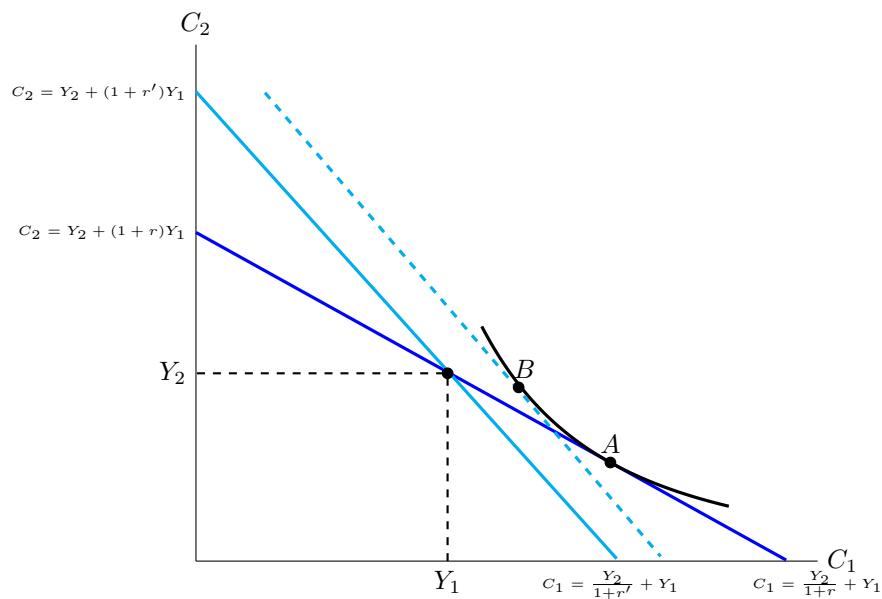
Solution: A borrower will select a point where their initial consumption C_1 exceeds their initial income Y_1 :



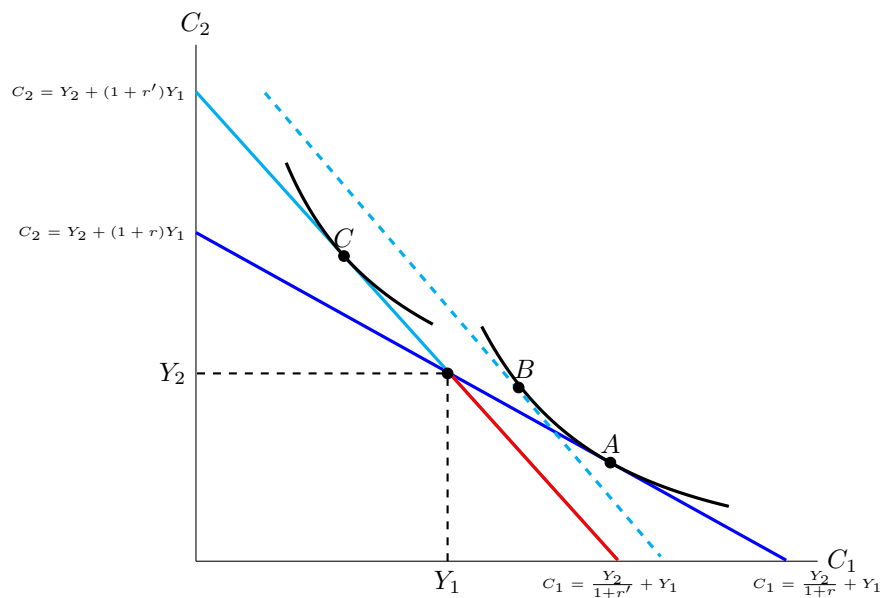
Following an increase in interest rate $r' > r$, the budget line pivots around the endowment point:



Just as we have learnt before, the Hicksian substitution effect is demonstrated by restoring the individual to the original utility curve but with the new price ratio:



Geometrically, the individual will select a point which gives them a higher utility compared to their initial point. For example, they would choose point C :

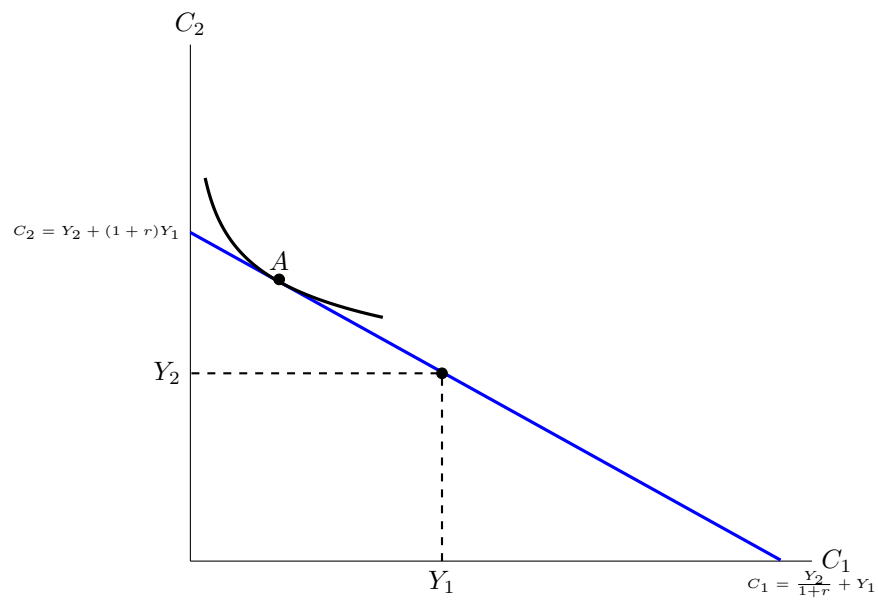


We can see that if the individual selects any point on the red line, they are choosing to continue as a borrower. However, this offers a lower utility compared to their initial position.

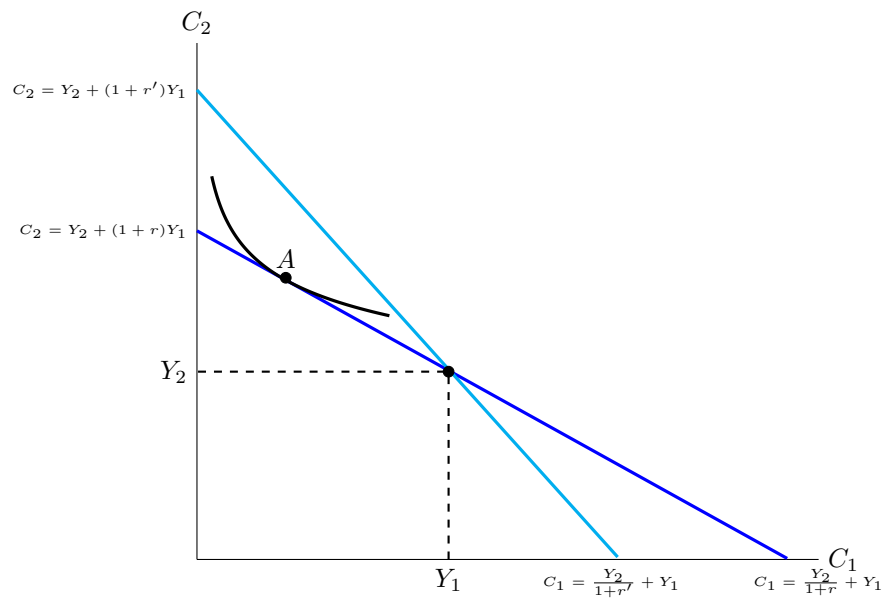
Intuitively, when interest rates increase, a borrower becomes poorer. This is because it is more expensive to take on debt. Therefore, the increase in r will force the consumer to borrow less or to become a saver, such as at point C .

Example 2.B: Now consider the same scenario of a rise in interest rate but the individual begins as a saver.

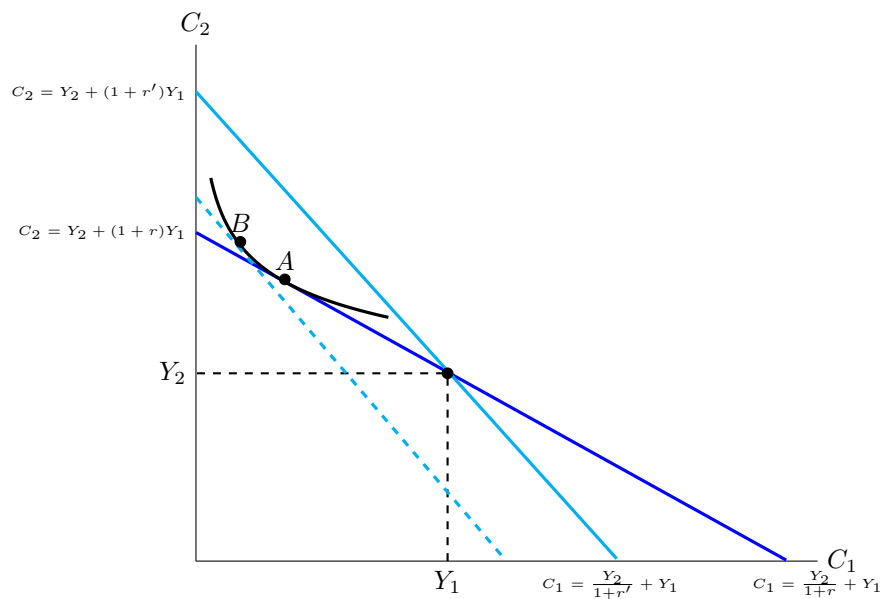
Solution: We begin with the individual choosing a point where $C_1 < Y_1$:



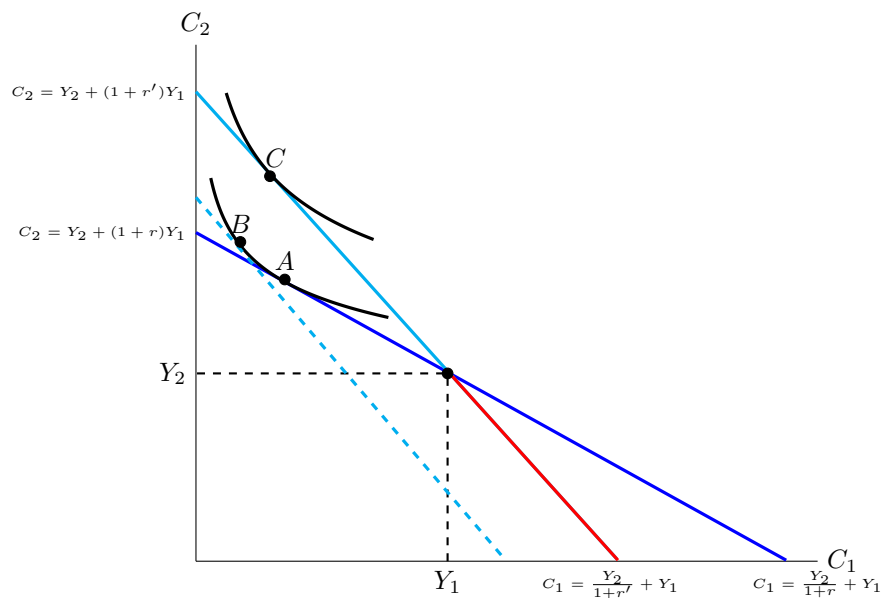
Following a rise in interest rates $r' > r$, the budget line rotates around the endowment point:



The Hicksian substitution effect would restore the individual to the original utility curve but with at the new price:



Just as the previous case, the individual can select any point to the left of the endowment position as it offers a higher utility to them. For example, they will select point C :



They will not select a point on the red line, as it offers a lower utility position compared to their initial consumption decision.

Intuitively, a rise in interest rate makes the saver richer (since they benefit from getting more money from their savings). However, the increase in income can be large enough that it would encourage them to spend more.

Essentially, it is theoretically ambiguous where C will lie (although it will certainly be to the left of the endowment point). Notice that unlike the previous case, the substitution and income effect go in opposite directions. If the net substitution effect becomes negative, then it implies that C_1 is an inferior good.

4 Problem Set

1) A consumer has the following Cobb–Douglas utility function:

$$u(x, y) = x^\alpha y^\beta$$

where $\alpha, \beta > 0$. The price of x is normalised to 1 and the price of y is p . The consumer's income is M . Derive the demand functions for x and y .

2) Using the assumptions discussed in the chapter, explain why indifference curves of an agent cannot cross.

3) Consider a consumer's maximisation problem:

$$u(x, y) = x^2 + y^2 \quad \text{s.t.} \quad P_x x + P_y y = M$$

- (a) Derive the tangency condition using the Lagrangian and illustrate an interior solution on the graph.
- (b) Explain why this utility function does not satisfy the standard assumptions (*Hint: Show that it does not have diminishing MRS.*).

4) Compute the MRS for each utility function:

- (a) $u(x, y) = \sqrt{xy}$
- (b) $u(x, y) = \ln x + \ln y$
- (c) $u(x, y) = 20 + 3(x + y)^2$

5) A consumer has the below utility function:

$$u(x, y) = \min\{\alpha x, \beta y\},$$

where $\alpha, \beta > 0$. Moreover, let $P_x X = 1$ and $P_y = W$, with the consumer's income set at M .

- (a) Derive the Marshallian demand function for x and y . *Hint: The utility function implies that both goods are perfect complements to the consumer.*
- (b) Suppose that the price of x falls. Illustrate how this will have no substitution effect.

6) Consider a utility function where the goods are perfect substitutes:

$$u(x, y) = x + y$$

- (a) Let us say that $P_x = 4$ and $P_y = 2$. Illustrate how this leads to a corner solution. *Hint: Plot the budget line and indicate the bundle which maximises the consumer's utility.*
- (b) Suppose that $P'_x = 1$. Illustrate the new budget line and calculate the Equivalent Variation.

5 Solution

1) The consumer's problem is:

$$\max_{x,y} x^\alpha y^\beta \quad \text{s.t.} \quad x + py = M$$

Using the Lagrange multipliers method:

$$\mathcal{L} = u(x, y) + \lambda(M - x - Py)$$

Deriving the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda P = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - x - Py = 0.$$

From the first two equations, we start by making y the subject:

$$\alpha x^{\alpha-1} y^\beta = \frac{\beta x^\alpha y^{\beta-1}}{P}$$

$$\frac{y^\beta}{y^{\beta-1}} = \frac{\beta}{\alpha} \frac{1}{P} \frac{x^\alpha}{x^{\alpha-1}}$$

$$y = \frac{\beta}{\alpha} \frac{1}{P} x.$$

To find the optimised value of x , we substitute y into the budget constraint and make x the subject:

$$M = x + Py$$

$$M = x + P \frac{1}{P} \frac{\beta}{\alpha} x$$

$$M = \frac{\alpha x + \beta x}{\alpha}$$

$$x^* = \frac{\alpha M}{\alpha + \beta}.$$

The optimised value of y is therefore:

$$y = \frac{\beta}{\alpha} x$$

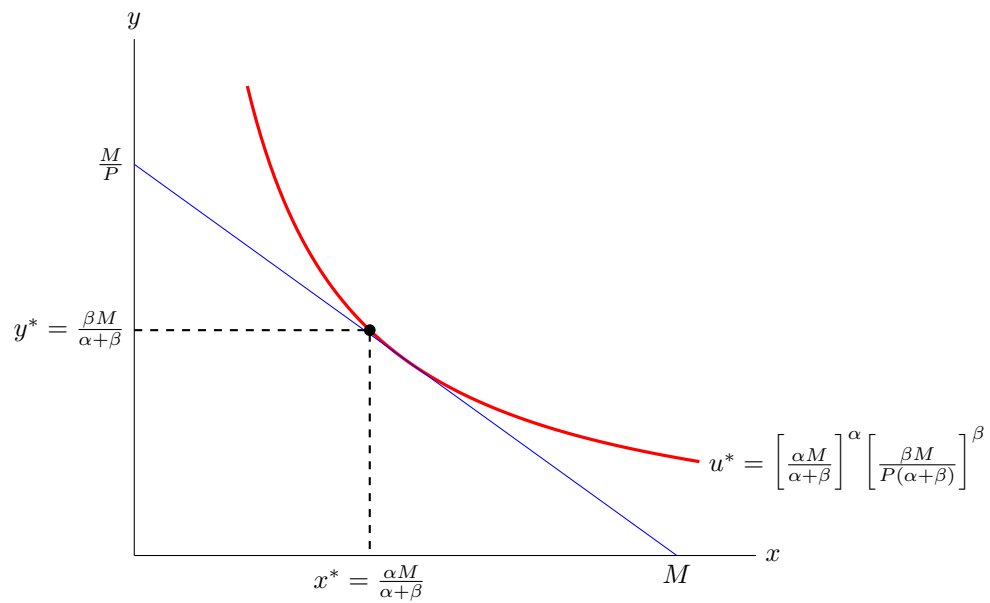
$$y = \frac{1}{P} \frac{\beta}{\alpha} \frac{\alpha M}{\alpha + \beta}$$

$$y^* = \frac{\beta M}{P(\alpha + \beta)}.$$

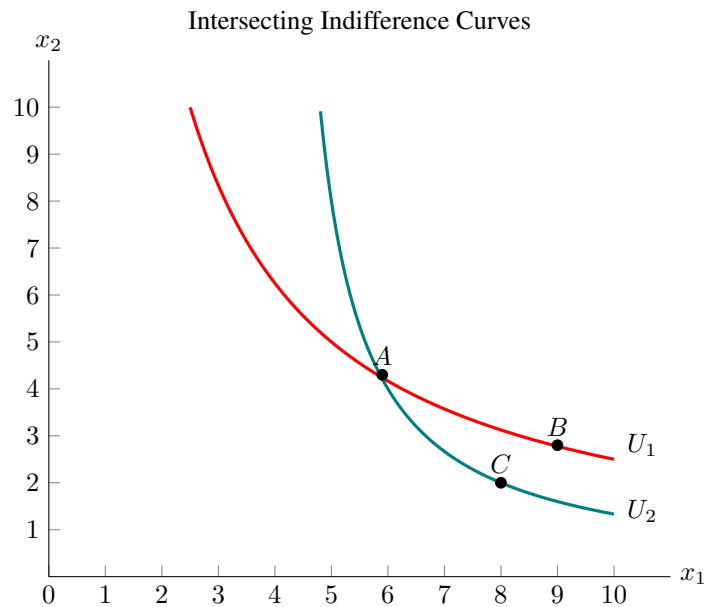
Which means that the value of utility at the optimum is:

$$u^* = \left[\frac{\alpha M}{\alpha + \beta} \right]^\alpha \left[\frac{\beta M}{P(\alpha + \beta)} \right]^\beta$$

Graphically, this can be shown as:



2) Consider the below indifference curves of a consumer:



Notice how both curves intersect at point A . If we examine the red line U_1 , this means that the consumer gets the same utility (and therefore indifferent) at point A and point B . Similarly, if you examine the green line U_2 , the consumer should be indifferent between A and C .

By transitivity, the result is that consumer should also be indifferent between both B and C . But this should not be possible, since B offers **more of both goods** compared to point C . By monotonicity, the consumer **should prefer** B over C and not be indifferent.

Therefore, a consumer cannot have intersecting indifference curves as it violates our assumptions about preferences.

3) (a) We can derive the Lagrangian as:

$$\mathcal{L} = x^2 + y^2 + \lambda(M - P_x x - P_y y)$$

The first order conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda P_x = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y - \lambda P_y = 0$$

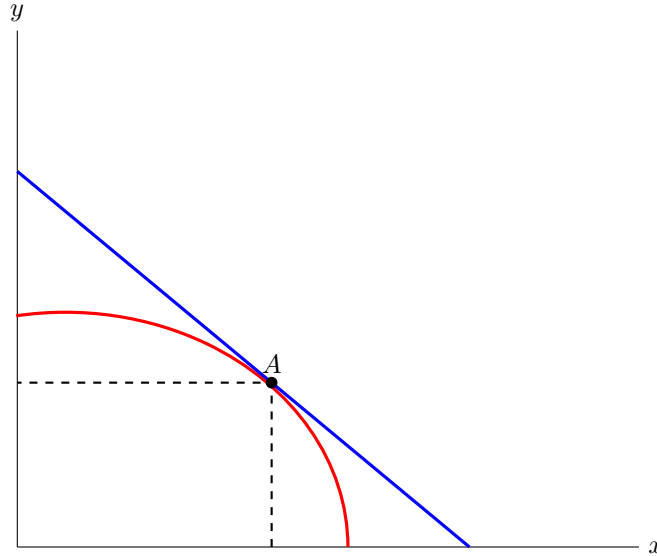
$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - P_x x - P_y y = 0$$

Equating the first two equations:

$$\frac{x}{y} = \frac{P_x}{P_y},$$

which is the point of tangency $MRS = \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$.

An interior solution can be shown as the point of tangency:



(b) This utility function does not satisfy the standard assumptions, particularly the convexity property. For your module, it is sufficient to simply explain that the MRS is not diminishing:

$$MRS = \frac{x}{y},$$

since as x increases, the MRS increases. This means that the second order conditions are not satisfied and so point A is not an optimum.

Interested students can test the second order conditions using formally using matrix algebra. **Do note that this is beyond the scope of your module.**

The sufficient second order conditions of a maximum in a constrained optimisation function requires that the symmetric matrix of the Lagrangian's second derivatives are negative definite.

The Hessian can be shown as:

$$A = \begin{bmatrix} \mathcal{L}_{11} & \mathcal{L}_{12} & \mathcal{L}_{13} \\ \mathcal{L}_{21} & \mathcal{L}_{22} & \mathcal{L}_{23} \\ \mathcal{L}_{31} & \mathcal{L}_{32} & \mathcal{L}_{33} \end{bmatrix}$$

From the Lagrangian equation defined earlier, this would mean:

$$A = \begin{bmatrix} u_{11} & u_{12} & -P_x \\ u_{12} & u_{22} & -P_y \\ -P_x & -P_y & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -P_x \\ 0 & 2 & -P_y \\ -P_x & -P_y & 0 \end{bmatrix}$$

In order to check if the bordered Hessian is negative definite, we need to show that the determinants of the first and second principle minors have opposing signs:

$$A_1 = \begin{bmatrix} 2 & -P_x \\ -P_x & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 & 0 & -P_x \\ 0 & 2 & -P_y \\ -P_x & -P_y & 0 \end{bmatrix}$$

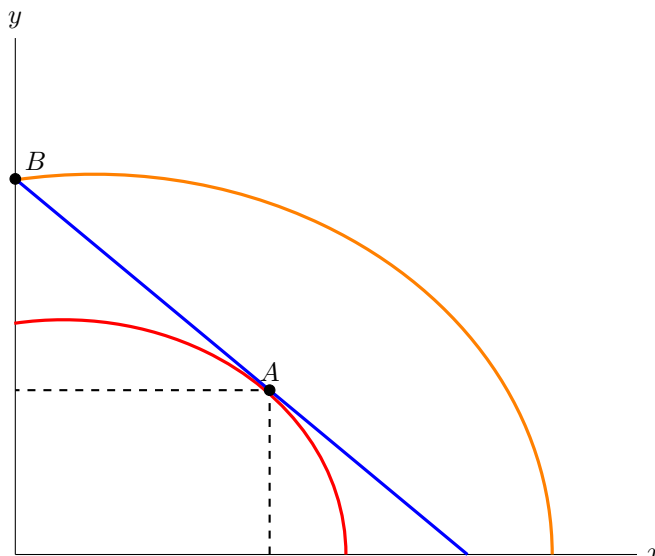
The determinants are

$$|A_1| = 2 \cdot 0 - (-P_x \cdot -P_x) < 0$$

$$|A_2| = 2(2 \cdot 0 - (P_y \cdot P_y)) - 0(0 \cdot 0 - (-P_y \cdot -P_x)) + (-P_x)(0 \cdot -P_y - (2 \cdot -P_y)) < 0$$

Since they both have the same signs, the sufficient conditions for maximisation fails. This means that the interior tangency solution is not an optimum (it is actually a minimum).

The consumer will therefore only have a corner solution that maximises their utility, such as the vertical or horizontal intercept point:



Intuitively, the customer simply exhausts their budget on the good that is cheapest to purchase.

4) The MRS is given by:

$$MRS = \frac{MU_x}{MU_y}$$

(a) For $u(x, y) = \sqrt{xy}$

$$\frac{\partial u}{\partial x} = \frac{\sqrt{y}}{2\sqrt{x}}$$

$$\frac{\partial u}{\partial y} = \frac{\sqrt{x}}{2\sqrt{y}}$$

The MRS is therefore:

$$MRS = \frac{\frac{\sqrt{y}}{2\sqrt{x}}}{\frac{\sqrt{x}}{2\sqrt{y}}} = \frac{\sqrt{y}}{2\sqrt{x}} \cdot \frac{2\sqrt{y}}{\sqrt{x}} = \frac{y}{x}$$

(b) For $u(x, y) = \ln x + \ln y$

$$\frac{\partial u}{\partial x} = \frac{1}{x}$$

$$\frac{\partial u}{\partial y} = \frac{1}{y}$$

The MRS is therefore:

$$MRS = \frac{\frac{1}{x}}{\frac{1}{y}} = \frac{y}{x}$$

(c) For $u(x, y) = 20 + 3(x + y)^2$

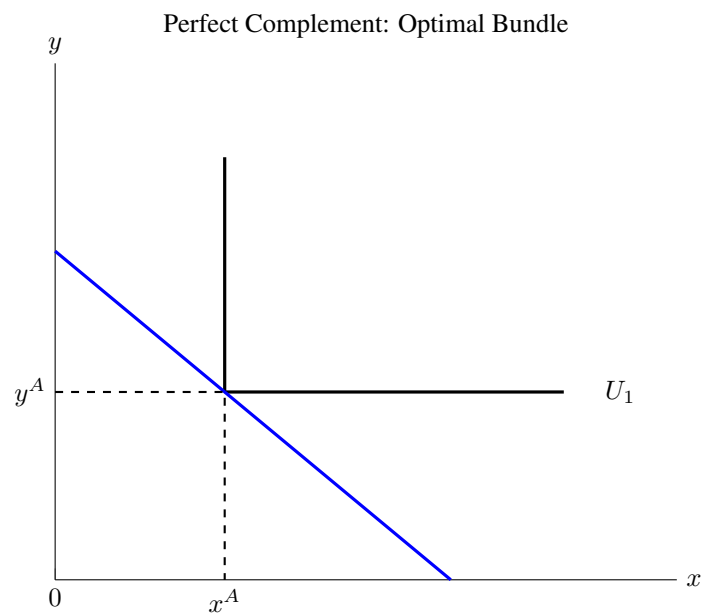
$$\frac{\partial u}{\partial x} = 6(x + y)$$

$$\frac{\partial u}{\partial y} = 6(x + y)$$

The MRS is therefore:

$$MRS = \frac{6(x + y)}{6(x + y)} = 1$$

5) Using the hint, we can illustrate the consumer's maximisation problem as:



(a) The consumer only gets utility from consuming x and y in fixed proportions. Therefore, the consumer will always choose a bundle such that:

$$\alpha x = \beta y$$

This means that:

$$y = \frac{\alpha}{\beta} x$$

We know that the budget constraint is:

$$x + Wy = M$$

$$x + W\left(\frac{\alpha}{\beta}x\right) = M$$

$$x\left(1 + W\frac{\alpha}{\beta}\right) = M$$

$$x = \frac{M}{1 + \frac{W\alpha}{\beta}}$$

$$x = \frac{M}{\frac{\beta + W\alpha}{\beta}}$$

$$x^* = \frac{M\beta}{\beta + W\alpha}$$

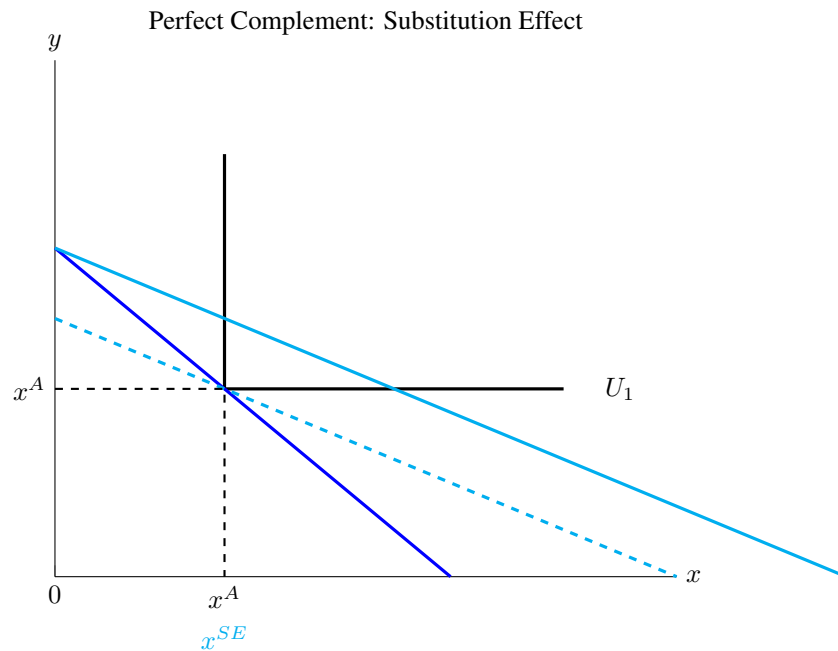
Substitute this into y :

$$y = \frac{\alpha}{\beta}x^*$$

$$y = \frac{\alpha}{\beta} \frac{M\beta}{\beta + W\alpha}$$

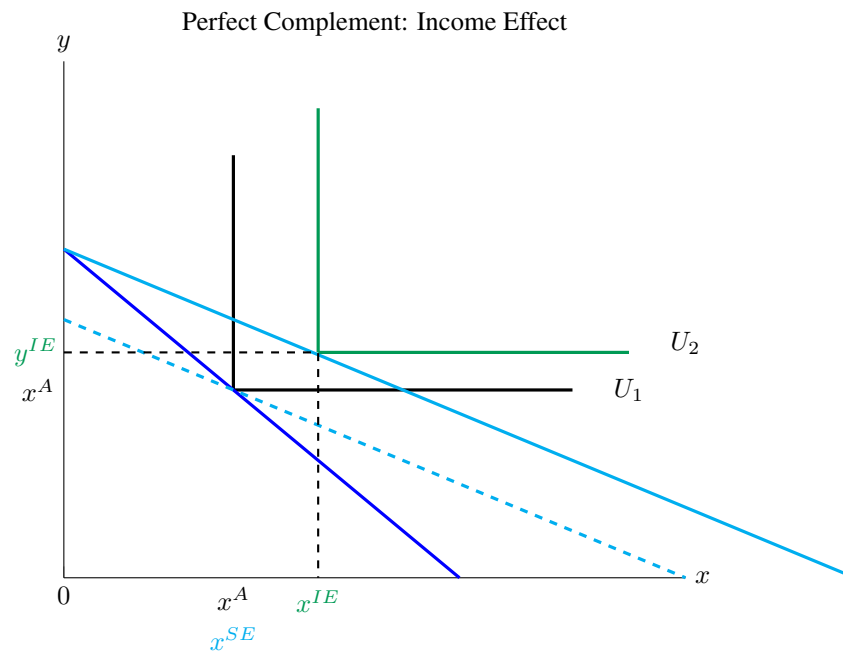
$$y^* = \frac{M\alpha}{\beta + W\alpha}$$

(b) When the price of x falls, the budget line rotates outwards. The Hicksian decomposition is illustrated when the budget line with the new price ratio is tangent to the old utility curve:



We can clearly see that this is only tangent at the kink. Therefore, the consumption bundle of the original point and the substitution point are equal.

In fact, there is only an income effect. By non-satiation, the consumer selects a consumption bundle that is on a higher utility curve:



6) You won't be able to solve this using the Lagrangian. It is easiest to see the solution when you plot it.

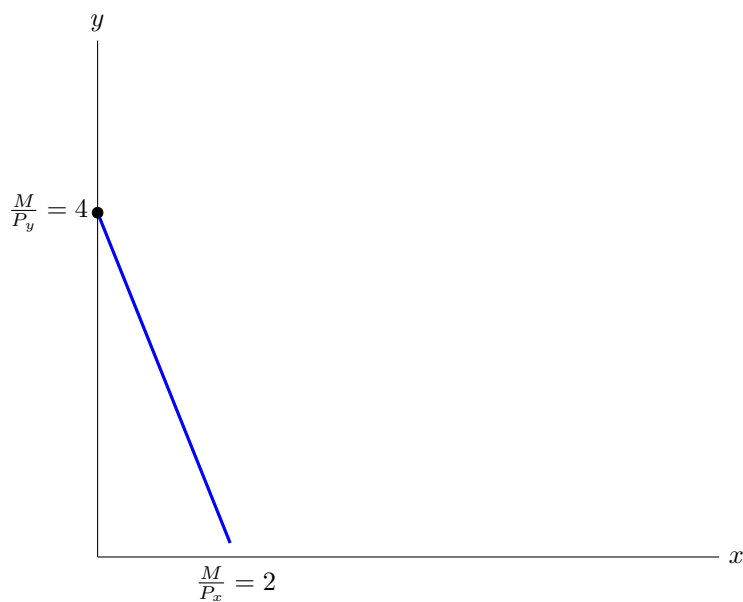
(a) In order to plot this in a graph where y is on the vertical axis and x is on the horizontal axis, we need to make y the subject of the budget line equation:

$$M = P_x X + P_y Y$$

$$8 = 4x - 2y$$

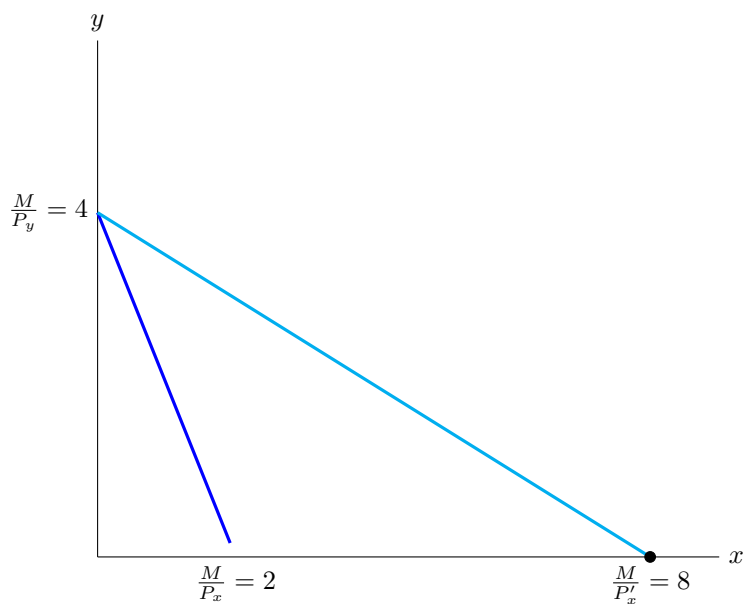
$$y = 4 - 2x$$

We can use this to join the vertical and horizontal intercepts:



The consumer considers the goods to be perfect substitutes, so they will either spend all their income on x or on y . Since good y is cheaper, then the consumer maximises their utility by selecting $y = 4$. This must mean that $u^* = 4$.

(b) Now consider if $P'_x = 1$. We can illustrate the new budget line as:



With this change, the consumer maximises their utility position by selecting the horizontal intercept point $x = 8$, which means that $u = 8$

The equivalent variation is the change in income that sets the bundle at the new utility position but using the original price.

The only thing to keep in mind is that the price we use in reflects the price of y . This is because we originally consumed only good y when the prices were at the original level.

Formally, we would say that:

$$\frac{M + EV}{P_y} = \frac{M}{P'_x}$$

$$\frac{8 + EV}{2} = \frac{8}{1}$$

$$EV = 8$$